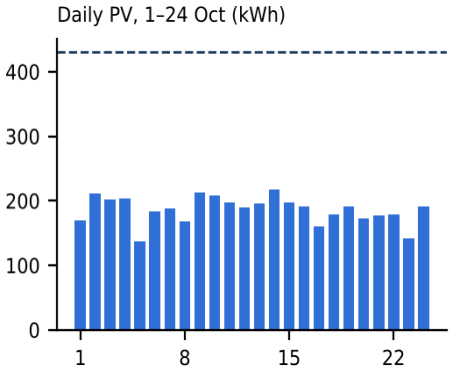
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

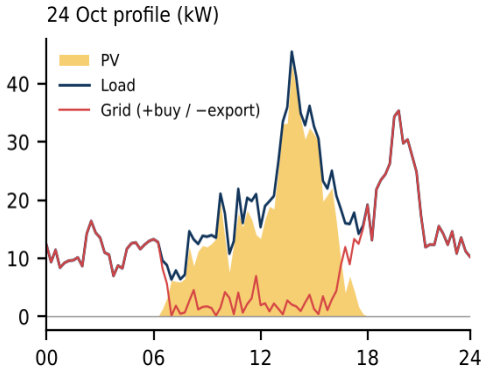
Food factory, central Thailand

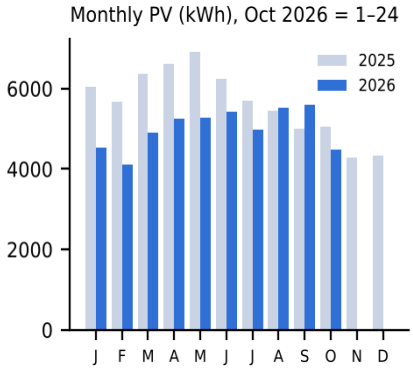
Upgrade opportunity

Zero-export curtailment – battery opportunity

100 kWp	4,465 kWh	1.86	10,363 kWh	5,898 kWh	1 kWh
PV · 100 kW inverter	PV, 1–24 Oct	kWh/kWp/day	Consumption	Bought from grid	Exported







What we found

• 100 kWp on two 50 kW inverters made 186 kWh a day (1.86 kWh/kWp). Output follows the factory load: the meter shows only 0.1–7 kW drawn from the grid at midday, while PV stays well below its capacity.

• On 24 October PV peaked at 43 kW when the array could have given 70–80 kW. At 4.3 kWh/kWp, about 240 kWh a day is being held back by the zero-export limit.

• The factory bought 5,898 kWh from the grid in 1–24 October, mostly in the evening and at night (25–35 kW from 18:00 to 21:00).

• Both inverters are healthy and evenly loaded, with phase voltages of 231 V.

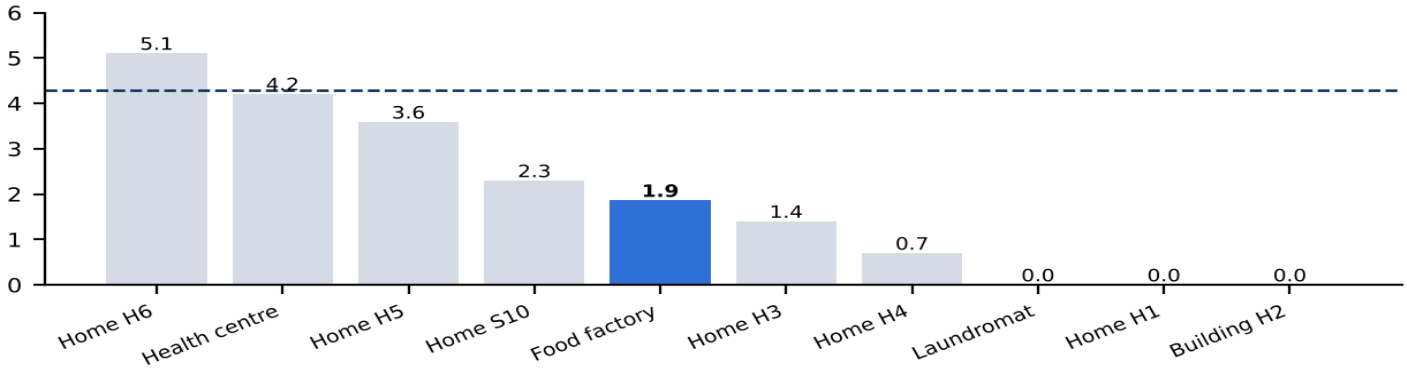
What to do

1. Offer a battery of about 150–200 kWh. It would store the curtailed midday energy and use it from 17:00 to 23:00, cutting grid purchase by about 180 kWh a day, worth around 24,000 THB a month.

2. Lower-cost step: move flexible loads (boilers, pumps, cold rooms, cleaning) to 10:00–14:00.

3. Check whether the factory can apply for PEA export, so surplus can be sold instead of thrown away.

How this plant compares (kWh/kWp per day, 1–24 Oct)



Rank 5 of 10 Huawei plants. Dashed line = expected 4.3 kWh/kWp per day.